PocketQuake

From a catalog CSV to a relocation notebook in one command

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What is PocketQuake?

A thin orchestrator that takes a list of earthquakes and produces a relocation + focal-mechanism notebook. It glues two specialised submodules:

- **seismoseo/necis-downloader** — KMA NECIS waveform fetcher (Playwright + the FTP staging server).
- **seismoseo/korea-cluster-relocation** — the picking + HypoInverse + HypoDD + SKHASH pipeline.

One command, end to end:

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju
```

Output: an executed Jupyter notebook with maps, depth sections, and beachball diagrams for every event you fed in.

Workflow at a glance

catalog.csv

$\downarrow$ *scaffold cluster dir under the eq-cycle submodule*

kma_waveforms/ (NECIS download via Playwright)

$\downarrow$ *eq-cycle pipeline*

PhaseNet+ picking → HypoInverse → ph2dt → dt.ct → xcorr → dt.cc → SKHASH

$\downarrow$

03_results_<slug>.ipynb – map, depth sections, beachballs

You write the catalog and run one command. PocketQuake takes care of dispatching to the right waveform source, registering the cluster, and executing the notebook.

Prerequisites

OS	Linux or macOS (bash wrapper; Windows uses WSL)
glibc	$\geq$ 2.28 — Ubuntu 20.04+, RHEL/Rocky 8+, Debian 10+. CentOS 7 / RHEL 7 are below the line (Playwright fails).
Python	3.10+ (pinned by `environment.yml`)
conda	Miniforge / Miniconda / Mambaforge
gcc/gfortran	$\geq$ 9.3 (for the compiled externals)
Disk	$\sim$10 GB (conda env + Chromium + waveforms)
Network	HTTPS to `github.com`, `code.usgs.gov`, `conda-forge`, `necis.kma.go.kr`

NECIS account (apply at `https://necis.kma.go.kr`, research / institutional accounts only; approval 1–5 days).
STP account is optional — only for pre-2020 waveforms.

Install [1/2] — Python environment

```
# 1. Clone with submodules
git clone --recurse-submodules https://github.com/seismoseo/PocketQuake.git
cd PocketQuake

# 2. Create the conda env (one command does everything Python-side)
conda env create -f environment.yml
conda activate pocketquake

# 3. Editable installs for PocketQuake itself + the NECIS submodule
pip install -e . -e external/necis-downloader

# 4. Playwright Chromium (one-time, ~200 MB)
playwright install chromium

# 5. Credentials
cp .env.example .env
#  then edit NECIS_USER, NECIS_PASS
```

Install [2/2] — External programs

PocketQuake's default chain calls these from the shell. None are pip-installable. Put each binary on `$PATH`:

- `hyp1.40` — HypoInverse (USGS Fortran)
- `ph2dt` + `hypoDD` — Waldhauser's HypoDD (Fortran)
- `mseed2sac` — IRIS converter (C)
- `libarchive` / `bsdtar` — already in `environment.yml`

Two research clones reachable by env var:

```
git clone https://github.com/AI4EPS/EQNet.git
echo "EQNET_DIR=$PWD/EQNet" >> ~/PocketQuake/.env

git clone https://code.usgs.gov/esc/SKHASH.git
echo "SKHASH_DIR=$PWD/SKHASH/SKHASH" >> ~/PocketQuake/.env
```

Full per-tool source + build instructions: **docs/EXTERNAL_TOOLS.md**.

Catalog format

A plain CSV with 10 columns, KST origin times:

```
Year,Month,Day,Hour,Minute,Second,Latitude,Longitude,Magnitude,Depth
2025,2,7,2,35,34,37.14,127.76,3.1,9
2025,2,7,2,54,38,37.14,127.76,1.4,6
2025,2,7,3,49,4,37.14,127.76,1.5,7
2025,2,8,10,13,23,37.14,127.76,1.6,7
```

- Origin time interpreted as KST (UTC + 9 h)
- PocketQuake auto-derives the cluster epicenter (catalog centroid) and the spatial bounds (bbox + 0.2°); override with `-epi`, `-bounds`
- Magnitude is informational — relocation does not use it
- Depth ≥ 0, measured in km

Run the chungju example

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju
```

The wrapper:

1. scaffolds a cluster dir under
   external/korea-cluster-relocation/chungju_cluster/
2. downloads NECIS waveforms (Playwright drives the JS portal, polls the staging queue, streams the FTP files)
3. runs PhaseNet+ → HypoInverse → HypoDD → SKHASH
4. builds + executes the results notebook

Useful flags:

- -source mixed — per-event STP+NECIS dispatch
- -picker stead — SeisBench PhaseNet (no EQNet needed)
- -mainshock <UTC> — Gwangyang-style mainshock treatment
- -cores N — cap xcorr workers (default 10)

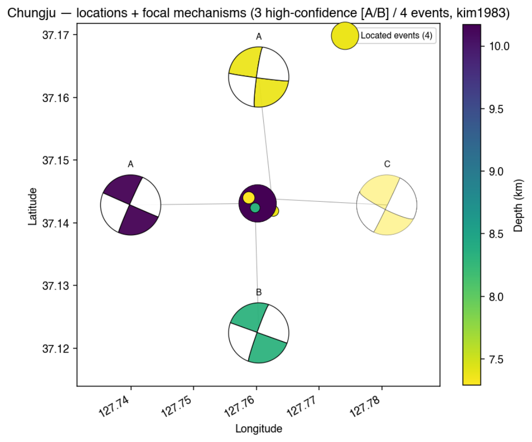
What you see while it runs

Foreground (`-fg`): full log to terminal.
Default (background): `nohup.out` in the working directory.

```
[pocketquake] scaffolding chungju_smoke_cluster ...
[necis] login OK; submitting request for 4 event(s)
[necis] request id 4f3a... status=P -> C, downloading 4 SAC bundles
[eqcycle] picking with phasenet_plus (~30s/event)
[eqcycle] hypoinverse: 4 events located
[eqcycle] xcorr 6 pairs (10 workers) ...
[eqcycle] dt.cc -> hypoDD reloc OK (4 events)
[eqcycle] focal_mechanism: 4 SKHASH solutions
[pocketquake] executing 03_results_chungju_smoke.ipynb
```

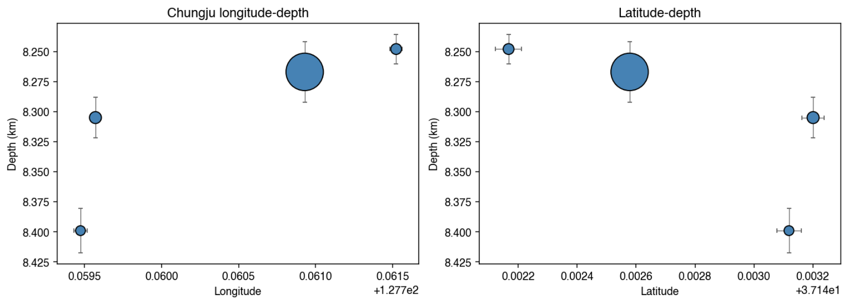
Wall-clock for chungju: $\sim$15 min on a 16-core box.

Output 1 — Relocated map + focal mechanisms



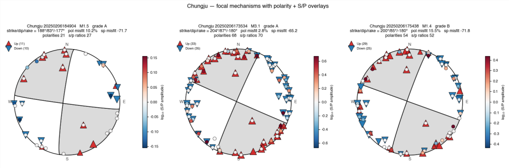
Four located events (depth-coloured dots) at $(37.142^\circ\mathrm{N}, 127.760^\circ\mathrm{E})$ with quality A/B/C SKHASH beachballs drawn on a leader-line ring. All four solutions converge on a near-vertical N–S right-lateral strike-slip plane.

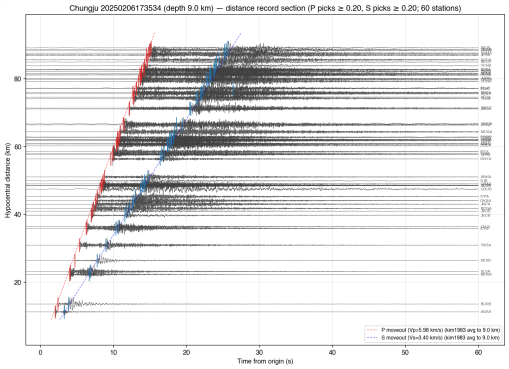
Output 2 — Depth sections



Three orthogonal cross-sections (E–W, N–S, along-strike) showing the depth distribution. HypoDD relative-relocation tightens the cluster from ∼1 km scatter (KMA bulletin) to ∼200 m.

Output 3 — Beachball gallery + record section





The notebook bundles a per-event beachball gallery (left) and a record-section view of the M3.1 mainshock with PhaseNet+ pick markers, first-motion polarities, and the SKHASH-inferred S/P amplitude ratios (right).

Run on your own catalog

1. Save your KMA-style CSV (the format from slide 6).
2. One command:

```
./pocketquake.sh ~/my_catalog.csv myswarm
```

3. Override the auto-derived epicenter / bounds if your catalog spans a wide region:

```
./pocketquake.sh ~/my_catalog.csv myswarm \
    --epi 35.46,128.43 \
    --bounds 35.3,35.65,128.25,128.65
```

4. Mainshock treatment (re-runs xcorr→dt.cc with the mainshock excluded, builds a `_main` notebook):

```
./pocketquake.sh ~/my_catalog.csv myswarm \
    --mainshock 20240912144719
```

Common pitfalls (`docs/TROUBLESHOOTING.md`)

`python interpreter not found` set `POCKETQUAKE_PYTHON=/path/to/python` in `.env` or put `python3` on `$PATH`.

`Permission denied (publickey)` use the HTTPS clone URL, not SSH.

`Submodule path '...' not registered` you forgot `-recurse-submodules`; fix with `git submodule update -init -recursive`.

`EQNET_DIR not set` PhaseNet+ is the default picker; either install EQNet (slide 5) or pass `-picker stead`.

`libstdc++ version not found` glibc/libstdc++ too old (CentOS 7); upgrade OS or run the NECIS download elsewhere.

`NECIS_USER` empty fill in `.env`.

Where to go next

- **Read the executed result notebook** — `external/korea-cluster-relocation/pipeline/notebooks/03_results_<slug>.ipynb`
- **Inspect the auto-derived configuration** — `external/korea-cluster-relocation/pipeline/clusters/<slug>.py`
- **Tune the run** — `./pocketquake.sh -help`
- **Other examples bundled** — `examples/sangju/`, `examples/chungju/`, the cluster modules registered in `korea-cluster-relocation/pipeline/clusters/`
- **Docs** — `docs/INSTALL.md`, `docs/EXTERNAL_TOOLS.md`, `docs/TROUBLESHOOTING.md`

`https://github.com/seismoseo/PocketQuake`

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